

Amith Sirisilla Jayadevgari

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PROFESSIONAL SUMMARY

Computer Science engineer specializing in deploying local Large Language Models (LLMs), Vision-Language Models (VLMs), and building high-performance, spatial-temporal Machine Learning (ML) pipelines. Proven track record engineering zero-idle-memory RAG pipelines, disk-backed hybrid retrieval engines, and real-time spatial compute engines with sub-second query latency. Adept at optimizing inference pipelines, vector databases, and 3D WebGL visualizations using PyTorch, DuckDB Spatial, Jina Embeddings, LanceDB, and Apache Arrow.

KEY PROJECTS

MedRAG (Clinical RAG Pipeline) | [GitHub](#) | *Python, LanceDB, LLM Router, FastAPI, NVIDIA NIM*

- Engineered an in-process, disk-backed hybrid retrieval pipeline combining **IVF-PQ vector search** and **Tantivy FTS** with **Reciprocal Rank Fusion (RRF)** in **LanceDB**, achieving **<150ms** query latency at **0MB idle RAM footprint**.
- Extracted layout-dense clinical tables and longitudinal EHR data at **94% parsing accuracy** using the **Chandra OCR 2** vision-language model, generating chunk-free embeddings via **Jina Embeddings v5** (32k context window).
- Engineered a dynamic synthesis router selecting between offline local LLMs (**Qwen 3.5**) and 3 cloud tiers based on complexity: **Ling 3.0 Flash** (fast lookups), **Laguna S 2.1** (>20k context), and **Nemotron 3 Ultra 550B** (complex clinical prognosis).

CityPlanner (Urban Optimization Agent) | [GitHub](#) | *Python, TypeScript, Gemma AI, DuckDB Spatial, Apache Arrow, FastAPI, Deck.gl*

- Automated multi-objective zoning planning by designing an AI agent system utilizing a local **Gemma** LLM for natural-language prompt processing and spatial feature engineering, cutting iterations by **65%**.
- Accelerated spatial database queries and relational vector intersections to **50k+ operations/sec** by designing a spatial indexing compute engine in **DuckDB Spatial**, cutting pipeline runtimes by **40%**.
- Eliminated dataset serialization latency by **70%** during real-time data ingestion, streaming structured tables via **Apache Arrow IPC** to a 3D WebGL Deck.gl dashboard rendering at **60 FPS**.

Project PEANUT (Environmental Forecasting) | [GitHub](#) | *TypeScript, Python, Next.js, GEE, Scikit-Learn, Tailwind, Supabase*

- Processed over **120GB of multispectral satellite imagery** via the **Google Earth Engine (GEE) API** for feature extraction of vegetation indices, scaling data pipeline analysis coverage by **3.5x**.
- Trained a multi-variable Polynomial Regression model in **Scikit-Learn** to predict resource deficits, achieving an **R² score of 0.88** and optimizing predictive accuracy over population projection datasets.
- Designed a spatial mapping dashboard serving predictions from supervised classification models, streaming structured GeoJSON layers via custom REST APIs.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, C, Bash

Frameworks & APIs: React, Next.js, Node.js, FastAPI, Streamlit, Deck.gl, Maplibre GL, Tailwind CSS

Machine Learning & LLMs: PyTorch, Hugging Face, Quantization (GGUF, AWQ), Hybrid Search (RRF), Cross-Encoder Rerankers, Ollama, LangChain, LlamaIndex, Scikit-Learn

NLP & Computer Vision: Vision-Language Models (VLMs), Document AI / OCR, Transformers, YOLOv8, OpenCV, Explainable AI (XAI), Grad-CAM

Databases & Spatial Computing: LanceDB (Vector DB), DuckDB (Spatial Extension), Apache Arrow (IPC / PyArrow), Google Earth Engine (GEE) API, Tantivy FTS, Supabase, PostgreSQL

Tools & Methodologies: Git, GitHub, Docker, Pytest, GitHub Actions (CI/CD), Linux Systems, Agile/Scrum, OOP

CERTIFICATIONS

Google Data Analytics Professional Certificate | *Coursera*

IBM Machine Learning Professional Certificate | *Coursera*

IBM Relational Database Administration (DBA) | *Coursera*

Artificial Intelligence Foundations: Thinking Machines | *LinkedIn Learning*

EDUCATION

Lovely Professional University

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.11/10.0

Jalandhar, Punjab

2022 – 2026

LEADERSHIP & ACHIEVEMENTS

- NGO Leadership & Volunteering:** Coordinated a team of 15 volunteers to execute outreach campaigns for child education, directly teaching and supporting 150+ underprivileged children.
- Community Organizing:** Led local public safety and child labor awareness initiatives, coordinating with community stakeholders to organize webinars with 300+ attendees.